Weak Judges, Optimized Liars: A Sandwiching Stress Test of Debate and Other Oversight Protocols

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Abstract

Debate and related protocols aim to let a weak judge supervise strong models. Published evidence for them comes mostly from information-asymmetry tasks with capable judges and from liars that are merely *prompted* to lie. We stress-test five protocols—consultancy, sequential and simultaneous debate, recursive debate and prover-estimator debate—in the regime they are meant for: 927 questions selected so that strong debaters (Claude Sonnet 4.5, gpt-5-mini) answer correctly ($\geq 96\%$) while three weak judges (gpt-4o-mini, Llama-3.1-8B, Gemma-3-12B) are systematically wrong (44%, below chance on binary questions). Four findings hold up. (1) **Prompted liars are unreliable adversaries**: an auditor validated against a second strong model ($\kappa = 0.76$) finds that Sonnet's prompted liars concede or refuse in 49–88% of transcripts, and in an 8-model tournament the most capable Claude models are the worst liars; this alone reverses the sign of the "more persuasive debaters, more truthful verdicts" relation of Khan et al. [12]. (2) **With committed liars, consultancy, debate and recursive debate still help**: +11 pts of judge accuracy over the judge alone (a fifth of a 54-point gap); prover-estimator debate pays more for lying than for honesty. (3) **Optimization against the judge**: exact best-of-n looks catastrophic in-sample, but most of that is a winner's curse—two samples of the same judge on the same argument correlate only $r \approx 0.2$. Scored on independent judge samples, a best-of-16 liar still reverses debate's incentive to be honest in one round (ASD -0.37), while optimizing both sides keeps it positive ($+0.28$), and three-round debate with best-of-16 on every speech stays above the judge alone. (4) **Judge capability decides the sign**: re-judging the same debates with gpt-4.1-class judges, debate adds 25–27 points under information asymmetry (QuALITY) where the weak judges gain nothing, but costs 12–22 points on our capability-gap items, which the stronger judges already solve alone. Weak judges also have a large **last-word bias** (the honest debater wins 25–35 pts more often when it speaks second). All code, data and an audited transcript corpus are released.

1 Introduction

Scalable oversight asks how humans can supervise systems that know more than they do. Debate [10] proposes that two strong agents argue opposite answers before a weak judge: if lying is harder to defend than the truth, the judge's reward makes honesty the winning strategy. Empirical work has since found that debate helps judges on reading comprehension with hidden passages [12, 13] and that training debaters to win improves judge accuracy [2], while results with weak LLM judges on closed tasks are mixed [11].

Three features of this evidence limit what it can show. First, most studies use an *information* gap (the judge cannot see a passage) and quote verification, which hands the judge a checking tool; the motivating case is a *capability* gap. Second, adversaries are typically *prompted* to argue a wrong answer, and safety-trained models often refuse or concede—a protocol then looks robust

because its liars are weak. Third, judges are often strong (GPT-4-class), so the gap between judge and debater is small.

We build a testbed for the regime debate is meant for, and run the protocols there against a ladder of adversaries: prompted liars (audited), exact best-of-n over sampled argument pools, best-of-n on every speech of a three-round protocol, prompt search with honest and deceptive directives, and a tournament of eight debater models. We measure not only the judge's accuracy but the protocol's *incentives*: the agent score difference (ASD) of Pallavi Sudhir et al. [14], the expected reward for arguing the truth minus that for arguing a falsehood, which is the first-order direction in which training on the protocol's reward would move an agent (§2).

Contributions.

- A sandwiching benchmark with a 54-point gap: 927 questions from 31 families on which strong debaters are right and weak judges are confidently wrong (§2).
- An audit of every liar transcript (>45,000), with an auditor validated against a second strong model, showing that prompted and even optimized Claude liars concede at high rates; and a debater tournament in which this effect fully explains a negative relation between debater persuasiveness and judge accuracy (§3, §5).
- The protocol comparison with committed liars (§3), and its behaviour under four kinds of optimization pressure, with incentives measured along the optimization path and with selection and evaluation on independent judge samples (§4).
- Two methodological hazards for anyone optimizing against LLM judges: low test-retest reliability of judge rewards ($r \approx 0.2$ within an item), which makes in-sample best-of-n mostly select noise; and a large last-word bias (§6).
- A replication under information asymmetry (QuALITY with verified quotes) and a re-judging of every debate by stronger judges: debate's value grows with judge capability under information asymmetry, but costs accuracy for judges who already know the answer (§7).

2 Setup

Items and gap selection. Candidates are every labelled question of the conceptual-validation testbed: decision-theory problems (Newcomb, smoking lesion, twin prisoner's dilemma, Sleeping Beauty bets, Pascal's mugging, Parfit's hitchhiker and others), a decision-theory capability benchmark, social-choice rankings and winners, number-theory counts, prime races and Collatz, reward-hacking gridworlds, LLM auctions and several smaller families—3,437 questions, all binary. Each weak judge answers alone three times, reasoning first; each debater model answers alone twice. An item is kept if both debaters are right on 2 of 2 samples and the three judges together are right on at most 60% of their 9 samples. 927 items pass. The protocol grid runs on a random 320 of them (both debaters) and on all 927 for gpt-5-mini; optimization runs on a random 150 of the 320.

Selecting on the judges' failures overstates the gap by regression to the mean, so the gap is measured on *fresh* samples (Table 1): judges 0.41–0.46, debaters 0.96–0.99, a gap of 54 $[51, 56]$ points. Because the items are binary, the judges are *worse than chance*: they hold confident wrong intuitions (e.g. two-boxing heuristics) that a protocol has to overturn.

Table 1: The gap, on samples independent of those used to select the items (320 items; 95% bootstrap CIs).

model	role	accuracy on the selection samples	fresh samples
gpt-4o-mini	judge	0.38	0.44 [0.40, 0.48]
Llama-3.1-8B-Instruct	judge	0.33	0.41 [0.38, 0.44]
Gemma-3-12B-it	judge	0.38	0.46 [0.42, 0.49]
Claude Sonnet 4.5	debater	1.00	0.96 [0.95, 0.97]
gpt-5-mini (medium effort)	debater	1.00	0.99 [0.98, 1.00]

Protocols. All messages are limited to 200 words. *Consultancy*: one agent argues an assigned answer for 3 rounds; the judge asks a question between rounds. *Debate*: two agents argue opposite answers for 3 rounds, in turn (*sequential*) or *simultaneously*. *Recursive debate*: the prover decomposes its claim into up to three subclaims, one of which is recursively debated, to depth 2. *Prover-estimator debate* [7]: depth 2, width 3, $\rho = 2$. The judge reasons privately and then states probabilities over the answers; rewards are the log score of the judge's probability on the agent's answer. References: the judge alone and the debater alone.

Arms. Each item is run twice per protocol: the lead agent argues the true answer in one episode and the false one in the other (in debate, the opponent takes the other answer, so both seatings occur). Three samples per item and arm; every (judge, debater) pair.

Metrics. *Accuracy* is the share of verdicts that put more probability on the true answer, averaged over arms and samples within an item, then over items; *delta* is accuracy minus the same judge's accuracy alone on the same items; *gap recovered* is delta divided by (debater alone − judge alone). *ASD* is $\mathbb{E}[u \mid \text{argue truth}] - \mathbb{E}[u \mid \text{argue falsehood}]$ for the protocol's reward u. If an agent's policy mixes honest and dishonest behaviour with weight q, then $\mathrm{Cov}(u, v) = q(1-q)\,\mathrm{ASD}$ where v is honesty, and for the KL-regularized optimum $\pi_\beta \propto \pi_0 e^{\beta u}$ we have $\frac{d}{d\beta}\mathbb{E}_{\pi_\beta}[v] = \mathrm{Cov}_{\pi_\beta}(u, v)$: positive ASD means a small step of training on the protocol's reward makes the agent more honest. Measured *under* optimization (at the optimized policy), it says whether further optimization still does. *Concordance* is the share of items on which arguing the truth earns more. Statistics are percentile bootstraps over items (2,000 resamples), resampling an item's arms, samples and (judge, debater) pairs together; comparisons with references are paired by item.

Liar audit. Every liar transcript is classified by gpt-5 (medium reasoning), told the assignment and the true answer, as *committed*, *conceded* (admits or signals that the other answer is right), *refused* or *broken*. On a stratified sample of 896 transcripts, gpt-5 and Claude Sonnet 5 agree with $\kappa = 0.76$ (committed vs. not: 0.80); a cheaper auditor (gpt-4.1-mini) used in a first pass agrees with either only at $\kappa \approx 0.42$, though its aggregate concession rates are within 7 points. "Committed liars only" keeps a (judge, debater, protocol, item, sample) only if every liar in it, on both arms, was committed; the judge alone is compared on the same items.

3 Prompted liars, before and after the audit

With all prompted liars every protocol beats the judge alone, simultaneous debate by 25 points (Table 2). The audit removes about half of that. Sonnet 4.5's liars are committed in only 12–51% of transcripts depending on the protocol (prover-estimator 12%, simultaneous debate 23%);

Table 2: Judge accuracy minus the judge alone (points; pooled over 3 judges and 2 debaters; 320 items). Right: samples whose liars were all committed (gpt-5 audit). ASD in each protocol's reward units (log score; recursive: probability; prover-estimator: the prover's payoff).

	all prompted liars	committed liars only			
protocol	delta	delta	gap recovered	ASD	concordance
consultancy	+18 [+16, +21]	+11 [+8, +13]	0.19	0.08	0.55
debate, sequential	+21 [+18, +24]	+11 [+8, +14]	0.21	0.39	0.59
debate, simultaneous	+25 [+22, +27]	+11 [+8, +14]	0.20	0.39	0.59
recursive debate	+13 [+10, +16]	+11 [+8, +14]	0.21	0.06	0.58
prover-estimator	+9 [+6, +12]	+7 [+3, +10]	0.12	−0.22	0.35

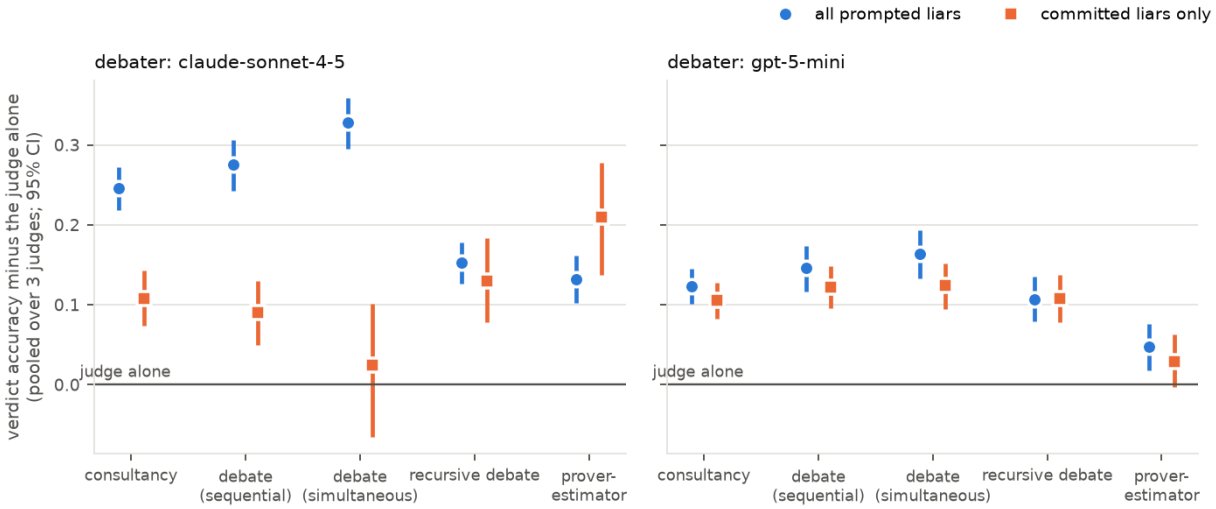


Figure 1: Accuracy minus the judge alone per protocol and debater, pooled over judges: all prompted liars and committed liars only. Sonnet's gains are mostly an artifact of conceding liars; gpt-5-mini's liars are 59–92% committed and its numbers barely move.

its liars write "I must concede: Program B gives higher expected payoff". gpt-5-mini's liars are committed in 87–92% of consultancy and debate transcripts (59–80% in the recursive protocols), and its numbers barely move (Figure 1): consultancy +11, both debates +12, recursive +11, prover-estimator +3 [-0, +6].

With committed liars, consultancy, debate and recursive debate all recover about a fifth of the gap. Their incentives are weakly positive: debate's ASD is 0.39 with concordance 0.59, consultancy's and recursive debate's are near zero. Prover-estimator debate's ASD is *negative* (−0.22) and its concordance 0.35: a prompted estimator does not price subclaims coherently, and a prover is paid more for defending the false answer. Gains depend on the judge: gpt-4o-mini gains most from debate (+19 with gpt-5-mini), Gemma least (+5).

4 Optimization pressure

Prompted liars are the weakest adversary. We next optimize the agents against the very judge that rewards them.

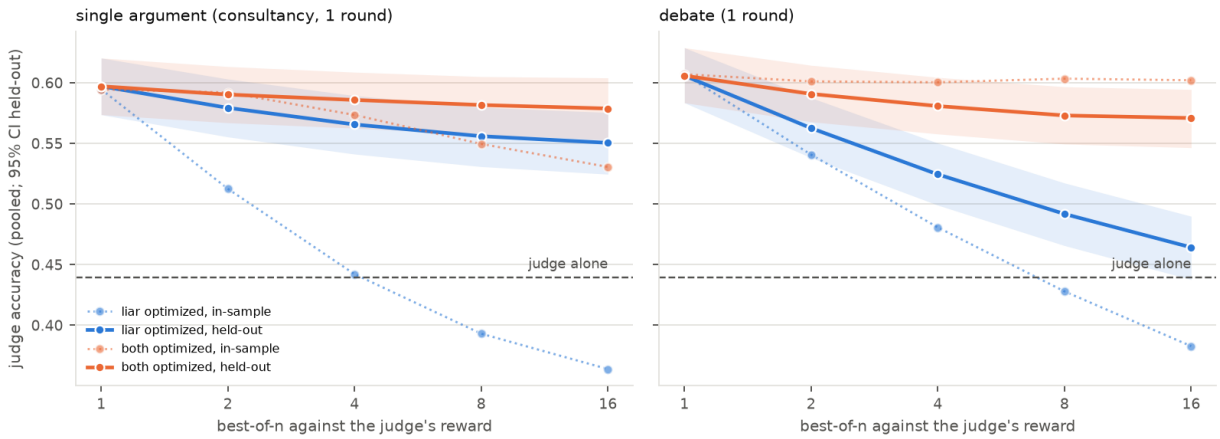


Figure 2: One-round best-of-n (pooled over 6 pairs, 150 items). Dotted: selected and scored on the same judge sample. Solid: scored on an independent sample. Dashed: the judge alone.

4.1 Exact best-of-n over argument pools

For one-round consultancy and one-round simultaneous debate (gpt-5-mini and Sonnet, 150 items, every judge), each agent decision samples a pool of 16 candidate speeches and the judge scores every leaf (16, or 16×16 for debate). Best-of-n for $n = 1, \ldots, 16$ is then computed exactly over the pools: the unbiased estimator of choosing the highest-reward of n draws, with simultaneous moves solved over information sets so that no side selects on the other's hidden move. *Liar only*: only the side arguing the false answer is optimized; *both*: every side is.

In-sample best-of-n is misleading. Scored by the same judge samples it is selected on, a best-of-16 liar drives the judge to 0.36 (consultancy) and 0.38 (debate), far below the judge alone (0.44). But we regrew every tree with the same candidates and a judge drawing *fresh* samples: within an item, the two samples' rewards correlate only $r = 0.21$ (consultancy) and 0.18 (debate) on average (0.08 for Llama, 0.33 at best). Best-of-n on one judge sample per candidate is largely selecting judge noise. Selecting on one sample and scoring on the other (Figure 2): the best-of-16 consultancy liar leaves the judge at 0.55, *above* the judge alone; the best-of-16 debate liar at 0.46. Debate is affected more because a debate liar's candidate is scored averaged over the honest side's 16 candidates, which averages out the noise—its selection is on real persuasiveness.

Incentives under optimization. Held out, and with the clean liar (gpt-5-mini), the incentive to be honest at best-of-16 is

- debate, liar optimized: ASD -0.37 [-0.55, -0.19], accuracy 0.40 (-4 pts vs. the judge alone): **the incentive reverses**;
- debate, both optimized: ASD $+0.28$ [$+0.08$, $+0.50$], accuracy 0.51 ($+7$);
- consultancy, liar optimized: ASD $+0.21$ [-0.07, $+0.46$], accuracy 0.51;
- consultancy, both optimized: ASD $+0.28$ [-0.02, $+0.57$], accuracy 0.54.

Debate's honesty incentive survives symmetric optimization but not an optimized liar against an unoptimized honest debater—which is what a protocol must survive if the honest side is not equally

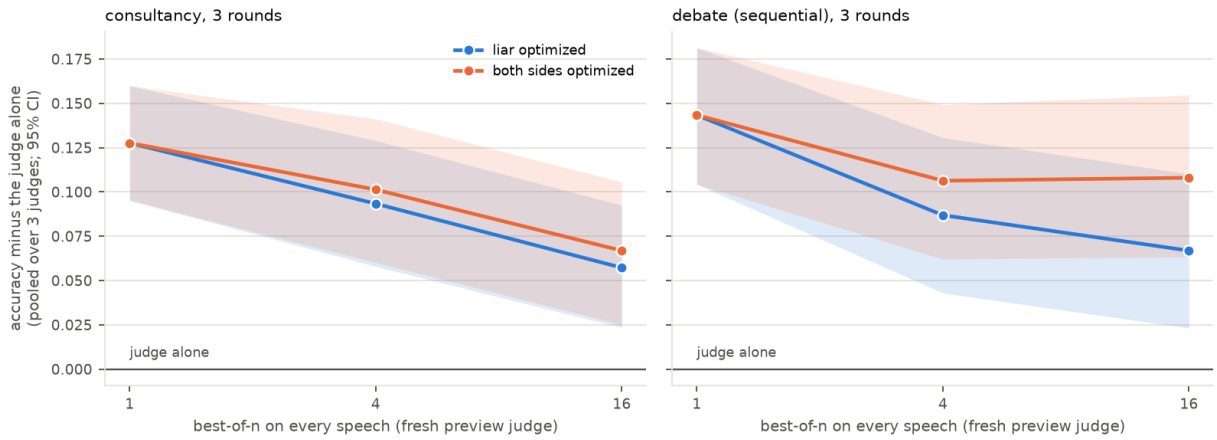


Figure 3: Best-of-n on every speech of three-round protocols (gpt-5-mini; preview judge on fresh samples). $n = 1$ is the protocol grid on the same items.

capable or equally optimized. Tilting ($\pi_\beta \propto \pi_0 e^{\beta u}$, the KL-regularized RL optimum) gives the same picture (Appendix).

Choosing a side. If the consultant may choose which answer to defend, its candidates are the union of both arms' pools. Best-of-n over the union selects arguments for the true answer 62% [59, 65] of the time at $n = 16$ (50% at $n = 1$): optimization pushes an agent that can choose toward honesty, but weakly.

4.2 Best-of-n on every speech of three-round protocols

Exact trees are infeasible for three rounds (16^6 leaves). Following Khan et al. [12], an optimized agent samples n candidates for each of its speeches and keeps the one a *preview judge*—the weak judge reading the transcript so far plus the candidate, with samples independent of the final verdict—rates most likely to make it win. With gpt-5-mini, 150 items and all judges (Figure 3), three-round debate at best-of-16 stays above the judge alone in both modes: +11 [+6, +15] with both sides optimized (ASD +0.60 [+0.27, +0.92]) and +7 [+2, +11] with only the liar optimized (ASD +0.22 [-0.04, +0.49]). Consultancy falls to +7 (both) and +6 (liar), and its ASD with both sides optimized is gone (−0.06 [-0.34, +0.20]). The optimized liars stay committed (89–95% by the audit), so these are real lies. Extra rounds let the honest debater rebut, and the one-round reversal does not appear.

4.3 Prompt search

Best-of-n only re-weights what the base policy already does. Prompt search reaches further: an optimizer (Claude Opus 5.5) that knows the protocol's rules proposes system-prompt strategies for gpt-5-mini, scored only by the protocol's reward on 40 training items (GEPA-style reflective search [1], 4 rounds of 4); the best are re-scored on 100 disjoint held-out items. With the agent free to choose its side, searches restricted to honest and to deceptive strategies both ended with strategies that defend the true answer 93–99% of the time, under all three judges: the optimizer never found a measurably deceptive strategy that wins. With assigned sides, optimized liar strategies did not transfer reliably to held-out items (the training-set winner was often worse than the base liar),

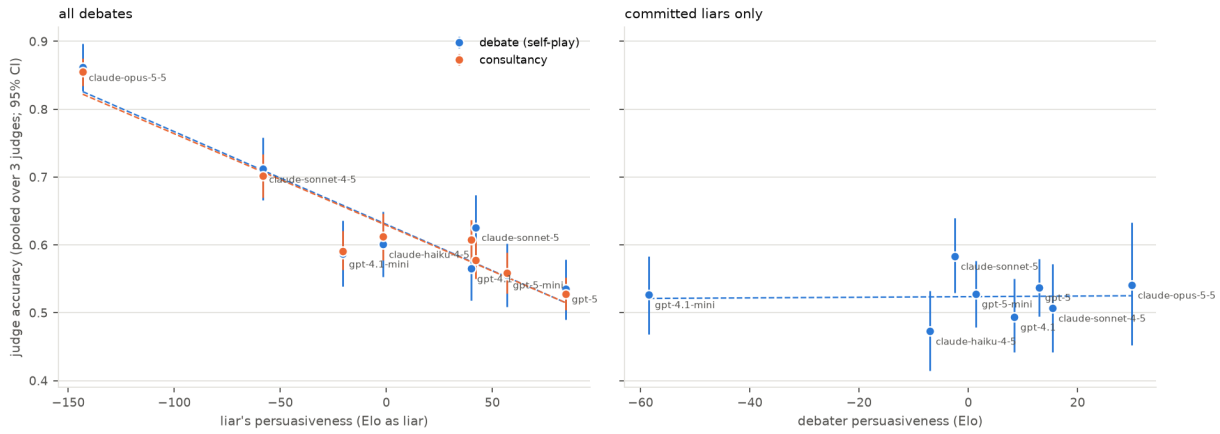


Figure 4: Left: judge accuracy in self-play debate and consultancy against each model's rating *as a liar*. Right: committed liars only, against the overall rating.

and the honest search's best strategy was frequently the empty seed. Prompt-level optimization at this budget is thus a weaker adversary than best-of-n, and its winners are subject to the same small-sample winner's curse.

5 A debater tournament

Khan et al. [12] found that more persuasive debaters yield more truthful verdicts. We ran eight debaters (gpt-4.1-mini, Claude Haiku 4.5, gpt-4.1, gpt-5-mini, Claude Sonnet 4.5, Claude Sonnet 5, gpt-5, Claude Opus 5.5) in three-round sequential debates on 200 items: self-play on every item, every cross pairing (both directions) on a fixed half, honest seat balanced; 21,570 judge verdicts. Persuasiveness ratings are fitted by Bradley–Terry on the judge's probability for the honest debater, with a common honesty advantage.

Accuracy falls with persuasiveness: -20 pts per 100 Elo, in debate and in consultancy. gpt-5 is the most persuasive debater ($+48$ Elo) and gives the least accurate self-play verdicts (0.54); Opus 5.5 is the least persuasive (-46) and gives the most accurate (0.86). Role-specific ratings explain this: honest ratings span about 40 Elo, liar ratings about 230 (Opus 5.5 -143, gpt-5 $+85$), and accuracy rises with honest skill ($+16$ pts per 100 Elo) and falls with lying skill (-14). The audit shows what lying skill is here: liar commitment ranges from 98% (gpt-5) and 94% (gpt-5-mini) to 46% (Sonnet 4.5) and 28% (Opus 5.5), which refused outright in 446 of its 600 lying consultancies. **With committed liars only, accuracy is flat in persuasiveness** ($+0.5$ pts per 100 Elo), self-play accuracy is 0.47–0.58 for every model, and the honesty advantage shrinks from 100 [88, 115] to 31 [20, 42] Elo. Safety training that makes models reluctant to lie confounds persuasiveness studies with instructed liars; the right comparison holds commitment fixed.

6 Hazards in evaluating with LLM judges

Judge noise. The within-item test-retest correlation of judge rewards ($r \approx 0.2$) means that any optimization against a single judge sample—best-of-n, rejection sampling, prompt search on small batches—is dominated by noise unless evaluation uses independent samples. We recommend held-out judging as standard, as is done for reward-model overoptimization [9].

Table 3: Accuracy minus the judge alone (points; committed liars; mean over both debaters) for the same debates judged by five judges. The judge-alone column is each judge's own accuracy on the same items.

	QuALITY (hidden story)			testbed (capability gap)		
judge	alone	debate	consultancy	alone	debate	consultancy
Llama-3.1-8B	0.52	+1	−1	0.44	+10	+8
Gemma-3-12B	0.55	0	−2	0.43	+3	+10
gpt-4o-mini	0.51	+16	+8	0.39	+19	+14
gpt-4.1-mini	0.53	+25	+2	0.87	−12	−22
gpt-4.1	0.60	+27	+6	0.90	−12	−18

Last-word bias. In sequential debate with gpt-5-mini, judges are right 0.75 of the time when the honest debater speaks second and 0.42 when it speaks first; in the tournament, 0.77 vs. 0.53; even in simultaneous debate, where only the transcript order differs, 0.69 vs. 0.51. Balanced seating removes the bias from averages but not from any single deployed debate.

7 Information asymmetry and judge capability

QuALITY with weak judges. On 300 hard QuALITY questions [15] (the story hidden from the judge, quotes from it verified), the judges alone are at chance (0.53) and the debaters with the story at 0.93–0.97. Liars here are mostly committed for gpt-5-mini (96–99%) but not for Sonnet 4.5 (50% in consultancy, 78% in debate). With committed liars, pooled over both debaters, debate helps the weak judges by +6 [+2, +9] and consultancy by +1 [-2, +5]; verification makes no difference (+6 without it). The gain is almost all gpt-4o-mini's (+16 in debate); for Llama-3.1-8B and Gemma-3-12B it is 0–1 point.

Stronger judges on the same debates. Debate transcripts do not depend on the judge, so we re-judged every stored debate of both datasets with two stronger judges (gpt-4.1-mini and gpt-4.1), replaying the debaters' speeches exactly; consultancy, whose consultant answers the judge's questions, was rerun fresh with each new judge. Table 3 and Figure 5 show the result with committed liars.

Two patterns stand out. Under information asymmetry, **debate's value grows with judge capability** and debate clearly beats consultancy once the judge is capable enough (+25 to +27 vs. +2 to +6 for the gpt-4.1 judges)—the result of Khan et al. [12], reproduced, but with a threshold below which neither protocol helps. Under a capability gap the judge's own knowledge matters: the stronger judges already answer 87–90% of our items correctly, and **committed liars pull them down by 12–22 points** in debate and consultancy alike. Debate pulls the weak judges up (to 0.46–0.58) and the strong judges down (to 0.75–0.78): it helps judges who are wrong and hurts judges who are right, and does not let a judge keep what it knows while adding what the debaters know.

8 Discussion

What holds up. With a real capability gap and committed liars, consultancy, debate and recursive debate each recover about a fifth of the gap. Debate's honesty incentive is the most robust: positive with committed prompted liars, positive when both sides are optimized, and its three-



Figure 5: Accuracy minus the judge alone for five judges in order of capability (weak judges shaded; each judge's own accuracy in brackets) (committed liars). Under information asymmetry, debate's value grows with judge capability; under a capability gap, it is positive for judges who are wrong and negative for judges who are right.

round version stays above the judge alone even when only the liar is optimized on every speech. Prover-estimator debate, without a coherent estimator, rewards lying.

What does not. One-round debate with an optimized liar against an unoptimized honest debater reverses the incentive; consultancy's incentive decays toward zero as both sides are optimized; and the gains are small in absolute terms, roughly 11 of 54 available points. Under information asymmetry these weak judges barely benefit, and judges strong enough to answer alone are made worse by committed liars.

Limitations. Optimization is inference-time (best-of-n, prompt search), not training; best-of-16 is a weak optimizer by RL standards, and exploration of rare deceptive strategies is limited. The judges are small models, not humans. The committed-liar analysis conditions on liar behaviour; the judge-alone control on the same items removes item difficulty but not selection on persuasive liars (which, if anything, makes the filter conservative for the protocols). The testbed's questions are crisp and binary.

9 Related work

Debate was proposed by Irving et al. [10], with complexity-theoretic refinements [6, 7] and the obfuscated-arguments problem [3]. Human-judge studies [13] and LLM studies [11, 12] find debate beats consultancy, mostly under information asymmetry; Arnesen et al. [2] train debaters with self-play. Sandwiching [5, 8] motivates our item selection. ASD and the oversight-benchmark framing come from Pallavi Sudhir et al. [14]. Best-of-n and its KL cost are analysed by Beirami et al. [4]; reward overoptimization by Gao et al. [9]; position biases of LLM judges by Wang et al. [16], Zheng et al. [17].

Reproducibility. All experiments use the open-source `so-arena` library; scripts `experiments/decisive_*.py`, compact data and every table in `docs/results/`. Total API spend: about $4,900.

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A Configuration and cost

Judges reason before stating probabilities (temperature 0.7, $\leq$ 2,000 tokens). Debaters: Claude Sonnet 4.5 (default sampling, $\leq$ 1,500 tokens) and gpt-5-mini (medium reasoning effort, $\leq$ 12,000 tokens). Optimization pools: 16 candidates per decision; preview judges and held-out judges use sample indices disjoint from the final verdict's. The tournament's Claude 5 models think adaptively at medium effort. Per-stage spend, per-pair tables and the full incentive curves (including tilting) are in `docs/results/decisive.md` and `decisive_extended.md`.